

RESISTOR COLOUR CODING

Aim

To study and identify resistor values using the resistor colour coding method.

Introduction

A resistor is a passive electronic component used to limit or control the flow of electric current in a circuit. Since resistors are very small, numerical values cannot always be printed on them. Therefore, coloured bands are used to indicate their resistance value, tolerance, and sometimes temperature coefficient.

Resistor colour coding is an internationally accepted method for identifying resistor values. By reading the coloured bands from left to right, the resistance value can be determined easily.

Colour Code Table

Colour	Number	Multiplier
Black	0	$\times 10^0$
Brown	1	$\times 10^1$
Red	2	$\times 10^2$
Orange	3	$\times 10^3$
Yellow	4	$\times 10^4$
Green	5	$\times 10^5$
Blue	6	$\times 10^6$
Violet	7	$\times 10^7$
Grey	8	$\times 10^8$
White	9	$\times 10^9$

Tolerance Colours

- Gold = $\pm 5\%$
- Silver = $\pm 10\%$
- No Band = $\pm 20\%$

Working Principle

Most resistors have four colour bands.

- First Band → First digit
- Second Band → Second digit
- Third Band → Multiplier
- Fourth Band → Tolerance

The resistance value is calculated using:

Resistance = (First Digit × 10 + Second Digit) × Multiplier

Example

Consider a resistor with the colours:

Brown – Black – Red – Gold

- Brown = 1
- Black = 0
- Red = ×100
- Gold = ±5%

Therefore,

Resistance = $10 \times 100 = 1000 \, \Omega$

Hence the resistor value is $1 \, \text{k}\Omega \pm 5\%$.

Applications

1. Current limiting in circuits.
2. Voltage division networks.
3. Biasing transistors.
4. Pull-up and pull-down resistor circuits.
5. Protection of LEDs and other electronic components.

Advantages

- Easy identification of resistor values.
- Internationally standardized method.
- Suitable for small-sized resistors.
- Quick and reliable for circuit assembly.

Conclusion

The resistor colour coding system provides a simple and effective method for identifying resistor values and tolerances. Understanding colour codes is an essential skill for electronics and IoT projects, helping engineers and students select the correct resistor for circuit applications.